

Surgical Approach for Morbid Obesity (Bariatric/Metabolic Surgery)

The Big Picture

Obesity = the plague of the 21st century — a **disease**, not a lifestyle choice. Bariatric surgery manipulates stomach and/or small bowel to achieve weight loss and control obesity-related disease. Because of strong effects on diabetes and metabolic syndrome, it's now called **metabolic surgery** (very cost-effective; can even resolve type 2 diabetes).

Why Surgery Works

The body defends a high "set point" — after weight loss it pushes to regain. Surgery **resets** that set point. ~15–25% weight loss is maintained for up to 20 years → long-term survival benefit + better quality of life.

Who Qualifies (NICE guidance, CG189)

- BMI ≥ 40 → option for anyone.
- BMI **35–40** with a comorbidity → expedited assessment.
- BMI **30–34.9** with **recent-onset type 2 diabetes** → consider.
- **Asian** populations → consider at a **lower BMI** threshold.
- Surgery only after non-surgical measures fail; patient must commit to lifelong follow-up.

Obesity-Related Disease (why we bother)

Type 2 diabetes, hypertension, dyslipidaemia, **obstructive sleep apnoea**, arthritis, GORD, NAFLD/steatohepatitis, PCOS, depression, several cancers (esp. **endometrial**).

The Team (multidisciplinary — essential)

Bariatric physician, dietician, specialist nurse, mental-health professional, **bariatric surgeon**, anaesthetist, radiologist, exercise therapist. Best outcomes in **high-volume specialised units**.

The Operations

- **Sleeve gastrectomy** — stomach tubed, ~80% resected. Restrictive. Excess-weight-loss 50–60%, diabetes remission ~50%.
- **Gastric bypass (Roux-en-Y / one-anastomosis OAGB)** — small pouch + bypassed bowel. Restrictive + **malabsorptive**. EWL 60–80%, diabetes remission ~80%.
- **Gastric band** — adjustable silicone band + subcutaneous port. Restrictive only. EWL 45–50%, lowest remission (~20%); now less favoured.
- **BPD/DS & SADI-S** — strongly malabsorptive duodenal switch. Highest EWL 70–80%, diabetes remission ~80%; most nutritional risk.

Complications

- **Early** — anastomotic/staple-line **leak** (~0.1–1%), bleeding, **DVT/PE**, wound/access-port infection. *Tachycardia may be the only sign of a leak — low threshold for re-laparoscopy.*
- **Late** — GORD (esp. sleeve), internal hernia (bypass), malnutrition/long-limb issues, marginal ulcer, stricture; band-specific: slippage, erosion, intolerance, weight regain.

Follow-Up & Monitoring (lifelong)

- **Supplements** — multivitamin + minerals always; **B12, iron, calcium, vitamin D, folate**; fat-soluble vits (A, E, K), zinc, copper, selenium after malabsorptive ops.
- **Bloods** — FBC, U&E, **HbA1c**, LFTs, ferritin, B12, folate, vit D, Ca/PTH, lipids; at 3, 6, 12 months then annually.
- Watch for night blindness, fatigue, anaemia, metabolic bone disease.
- **Shared-care model** — surgeon + primary-care collaboration for life.

Walk-Away Points

- Bariatric = **metabolic** surgery: best treatment for obesity + type 2 diabetes.
- NICE: ≥ 40 , or **35–40 + comorbidity**, or **30–34.9 + new diabetes**; lower for Asians.
- Bypass/DS beat sleeve and band for weight loss + diabetes remission.
- **Leak** is the feared early complication — tachycardia = re-look.
- Lifelong vitamins + monitoring, MDT, high-volume unit.

Morbid Obesity Surgery — Revision Table

NICE criterion (CG189)	Threshold		
Anyone	BMI ≥ 40		
With comorbidity	BMI 35–40 (expedited)		
Recent-onset type 2 diabetes	BMI 30–34.9		
Asian populations	Lower BMI threshold		
Operation	Mechanism	EWL	Diabetes remission
Sleeve gastrectomy	Restrictive (~80% resected)	50–60%	~50%
Gastric bypass (RYGB/OAGB)	Restrictive + malabsorptive	60–80%	~80%
Gastric band	Restrictive only (adjustable)	45–50%	~20%
BPD/DS, SADI-S	Mainly malabsorptive	70–80%	~80%
Complications	Examples		
Early	Leak (0.1–1%) , bleeding, DVT/PE, infection — *tachycardia = re-laparoscopy*		
Late (bypass)	Internal hernia, marginal ulcer, malnutrition		
Late (sleeve)	GORD / oesophageal reflux		
Late (band)	Slippage, erosion, intolerance, weight regain		
Monitoring	Detail		
Supplements	Multivit, B12, iron, Ca, vit D, folate ; +A/E/K, Zn, Cu, Se after malabsorptive		
Bloods	FBC, U&E, HbA1c, LFT, ferritin, B12, folate, vit D, Ca/PTH, lipids		
Schedule	Baseline, 3/6/12 mo, then annually — lifelong		
Obesity comorbidities			
Metabolic	T2DM, HTN, dyslipidaemia, NAFLD		
Other	OSA, arthritis, GORD, PCOS, depression, endometrial cancer		
MDT	Surgeon, physician, dietician, nurse, mental-health, anaesthetist, radiologist, exercise therapist		

Bold = high-yield.